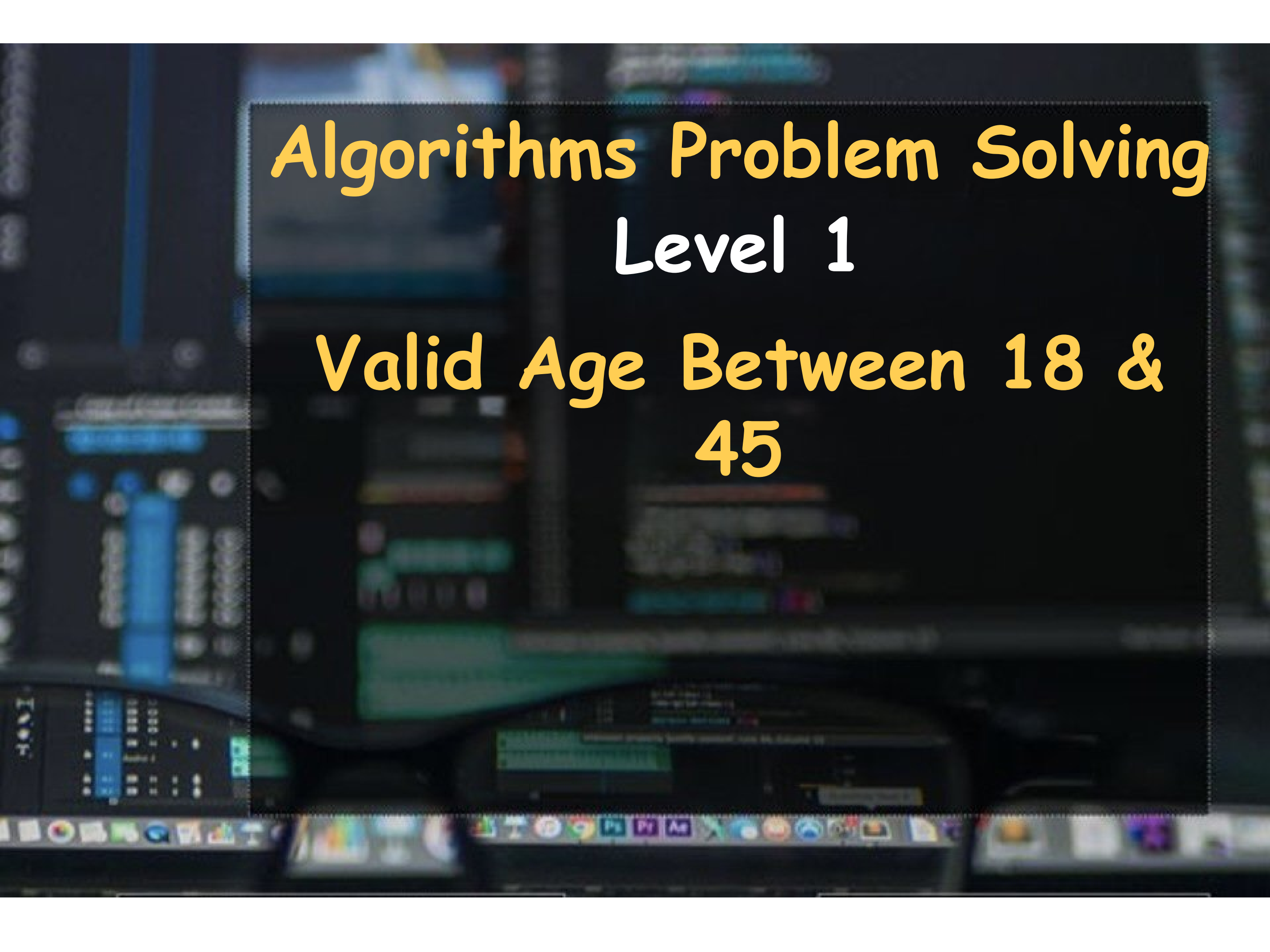
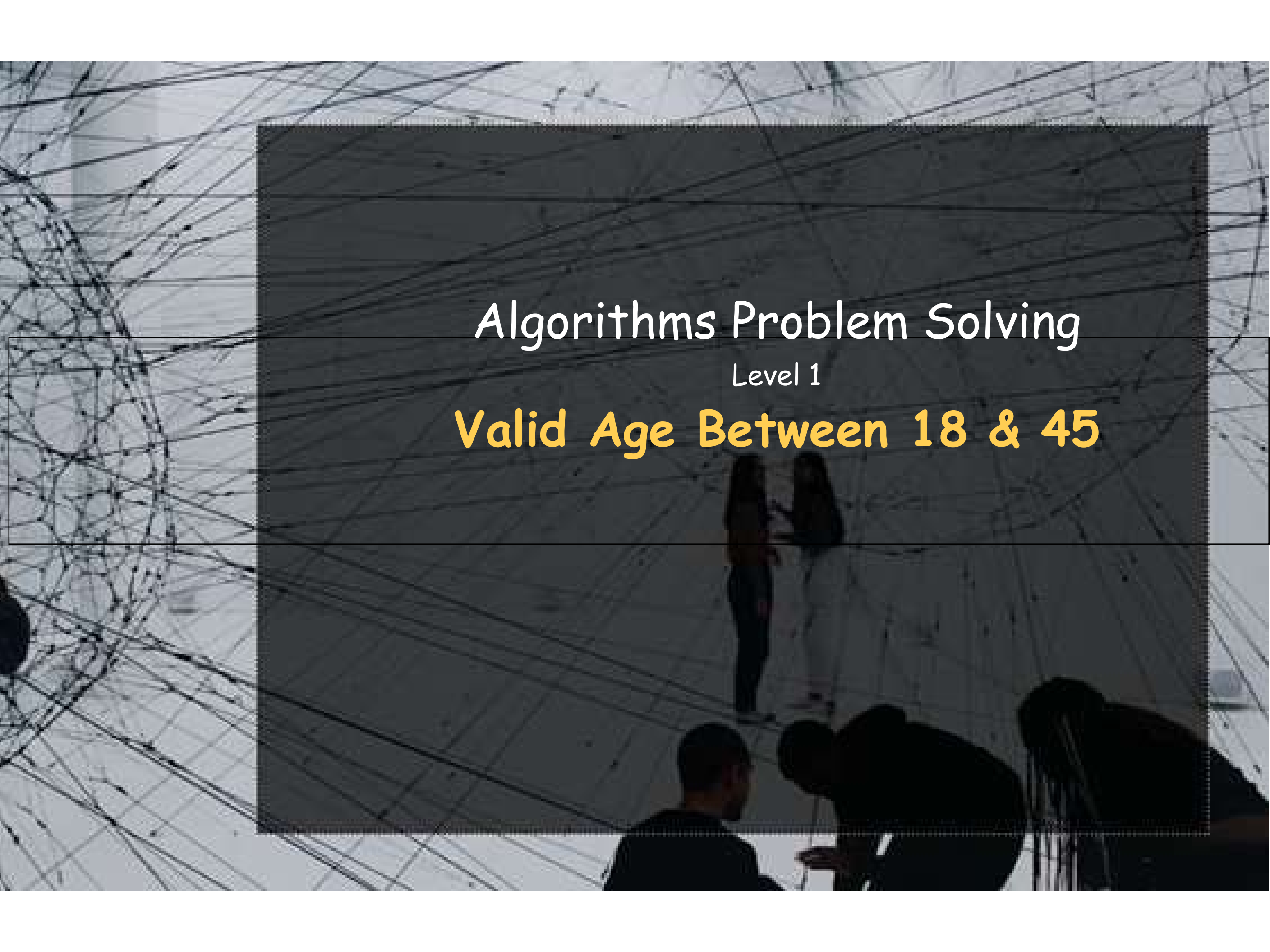




Algorithms Problem Solving

Level 1

Valid Age Between 18 &
45



Algorithms Problem Solving

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Valid Age Between 18 & 45

Definition

- ◆ This program reads the user's age and checks whether it is between 18 and 45.
- ◆ If the age is within this range, the program prints "Eligible for job".
- ◆ Otherwise, it prints "Not eligible for job".

Explanation

- ◆ ``readAge()`` asks the user to enter an age and returns it as an integer.
- ◆ ``validateNumberInRange()`` checks whether a number is within a specified range.
- ◆ ``Numb >= From`` checks that the number is not below the minimum.
- ◆ ``Numb <= To`` checks that the number is not above the maximum.
- ◆ The ``&&`` operator requires both conditions to be true.
- ◆ ``validateNumberInRange(age, 18, 45)`` checks whether the age is between 18 and 45, including both limits.
- ◆ ``printResult()`` prints the appropriate message based on the Boolean result.
- ◆ ``printResult(readAge())`` first reads the age, then passes the returned value to ``printResult()``.

Steps in main()

1. Read the user's age.
2. Pass the age to `validateNumberInRange()`.
3. Check if `age >= 18`.
4. Check if `age <= 45`.
5. Return true if both conditions are true.
6. Print "Eligible for job" if the result is true.
7. Otherwise, print "Not eligible for job".

💡 Example

◆ If the user enters:

Age: 25

◆ 25 is greater than or equal to 18 and less than or equal to 45.

➡ Result

25 is Eligible for job

Note

- ◆ The boundary values 18 and 45 are included because the conditions use `>=` and `<=`.
- ◆ The function `validateNumberInRange()` is reusable because `From` and `To` can be changed to check different ranges.

⊘ Warning

- ◆ The program checks the age range, but it does not validate whether the user entered a valid numeric value.



Comparison

Eligible

- ✓ Age is 18 or greater.
- ✓ Age is 45 or less.
- ✓ Both conditions must be true.

Not Eligible

- ✓ Age is below 18, or
- ✓ Age is above 45.
- ✓ At least one condition is false.

✦ Summary

- ✓ The program checks whether an age is between 18 and 45.
- ✓ `&&` requires both range conditions to be true.
- ✓ 18 and 45 are valid because the limits are inclusive.
- ✓ A Boolean function can return the result of a validation check.
- ✓ `printResult(readAge())` combines the functions by passing one function's return value to another.

Thank you 😊